VAIBHAV PANCHAL

Junior AI/ML Developer | GenAI & LLM Systems Engineer

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PROFESSIONAL SUMMARY

AI/ML Developer at TestGrid Labs Inc. building LLM-assisted test automation pipelines. M.E. in Software Engineering (LDCE, GTU, CGPA 8.21/10). Research in Computer Vision (YOLOv11 + ECA, +3.8% mAP50 on DIOR-R, under review) and Cybersecurity (hybrid IDS, 98.4% accuracy). Hands-on across the full ML lifecycle data engineering, model training, LLM/RAG systems, Flask deployment, and CI/CD. Core stack: PyTorch, TensorFlow, LangChain, LangGraph, FastAPI, Flask, AWS, Docker.

TECHNICAL SKILLS

Languages: Python (Advanced), SQL, JavaScript, C/C++, Java, HTML/CSS
Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace Transformers, XGBoost, LangChain, LangGraph, FastAPI, Flask, React.js, Node.js, Streamlit
AI/ML: Computer Vision, NLP, LLMs, RAG, Object Detection, Prompt Engineering, OpenAI API, Ollama, RAGAS, Langfuse, NeMo Guardrails
Cloud & Infrastructure: AWS (EC2), Docker, Nginx
Databases: SQLite, MongoDB, ChromaDB, Pinecone (Vector DB)
Tools & Practices: Git, Jupyter, MLflow, Postman, Selenium Grid, Bitbucket, REST API Design, CI/CD

WORK EXPERIENCE

Junior AI/ML Developer | TestGrid Labs Inc. | Navsari, Gujarat *May 2026 - Present*

- Engineered deep learning inference pipelines for automated UI regression detection, flagging anomalies before human review and reducing manual test triage time.
- Designed and deployed Flask-based model-serving endpoints integrated into the CI/CD pipeline for real-time ML inference at scale, with Nginx as reverse proxy for request routing and load distribution.
- Prototyped an LLM-assisted test case generation module using the OpenAI API, reducing repetitive test authoring effort across the QA team.

Trainee Associate Software Developer | Techreale Pvt. Ltd. | Valsad, Gujarat *Jan 2024 - Apr 2024*

- Delivered production features for the Famyll healthcare platform (React.js, Node.js, MongoDB) across claim management, auth flows, and real-time communication; reduced API response bugs by refactoring async integration patterns across 3 sprint cycles.

Summer Intern, Frontend Development (React) | INFOLABZ IT Services Pvt. Ltd. | Ahmedabad, Gujarat *Jul 2023*

- Developed and optimized React.js dashboard components with async API integration; improved rendering performance through targeted component-level refactoring.

PROJECTS

MESH Support — Multi-Agent Customer Support Platform *2026*

Python | LangGraph | FastAPI | MongoDB | NeMo Guardrails | Langfuse | Pytest

- Architected a LangGraph Supervisor Agent routing queries to specialized Billing, Technical, and Refund sub-agents with isolated prompts, tools, and memory, backed by FastAPI, MongoDB, and Groq's gpt-oss-120b.
- Integrated NeMo Guardrails input/output rails achieving 100% precision/recall (0% false-positive rate) on a labeled safety set (~1.8s overhead); delivered 100% end-to-end accuracy across routing, tool-selection, and guardrail evaluation (23/23 tests passing) with ~5.4s median guarded response time.

RAG System with Evaluation Pipeline *2026*

Python | FastAPI | React | Pinecone | Langfuse | RAGAS | BM25 | Groq

- Built hybrid retrieval combining Pinecone (MiniLM 384-d) with in-memory BM25 via Reciprocal Rank Fusion (k=60, α=0.6), enabling keyword-only fallback when the vector store is unavailable.
- Created a 45-question RAGAS golden set (faithfulness, answer relevancy, context precision) with a FastAPI evaluation endpoint and React dashboard; added prompt-injection filtering and deployed with API auth, non-root containers, and CI checks.

EDUCATION

M.E. — Software Engineering, GTU (L.D. College of Engineering, Ahmedabad) *2024 - 2026 | CGPA: 8.21/10*

B.E. — Computer Engineering, GTU (GDEC, Navsari) *2020 - 2024 | CGPA: 8.11/10*

ACHIEVEMENTS & CERTIFICATIONS

- Evaluate Generative AI Applications — Microsoft Learn: evaluation frameworks and metrics for Azure, Foundry Tools, and Content Safety (2026).
- Microsoft Learn Challenge | Ignite Edition: Build Trustworthy AI Solutions on Microsoft Azure (2025); Introduction to & Advanced Model Context Protocol — Anthropic (2026).
- SSIP Hackathon 2022 — automated smart attendance system, selected among top teams from Gujarat.